DESCENT OF THE BLOW-UP SQUARE CONDITION FOR COMPLEXES IN THE ESSENTIAL IMAGE OF THE DERIVED PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both directions, including the induction-free vanishing step; every invoked input is named in the text.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The morphism of sites epsilon and its basic properties. Same chapter, Section section-compare, in particular epsilon^{-1} epsilon_* = id. Supplies the adjunction and the identity used throughout the proof.
- The canonical map attached to a blow-up square. Same chapter, Lemma lemma-c-square. Supplies the maps c^{K'}_{X,Z,Y,E} whose being isomorphisms is characterized.
- Two out of three for the blow-up square. Same chapter, Lemma lemma-two-out-of-three-blow-up-square. Shows that the third term of a distinguished triangle satisfies the condition when two terms do.

Source reference. Original lemma 0F16: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6001–6001. Navigation pointer invoked at proof line 6688; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0EWP: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6046–6046. Navigation pointer invoked at proof

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line 6713; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original section 0EWK: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6177–6177. Navigation pointer invoked at proof line 6548; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Let $\mathcal{C}$ be a category. Let $\mathrm{Cov}(\mathcal{C}) \supset \mathrm{Cov}'(\mathcal{C})$ be two ways to endow $\mathcal{C}$ with the structure of a site. Denote τ the topology corresponding to $\mathrm{Cov}(\mathcal{C})$ and τ' the topology corresponding to $\mathrm{Cov}'(\mathcal{C})$. Then the identity functor on $\mathcal{C}$ defines a morphism of sites

$$\epsilon : \mathcal{C}_\tau \longrightarrow \mathcal{C}_{\tau'}$$

where ϵ_* is the identity functor on underlying presheaves and where ϵ^{-1} is the τ-sheafification of a τ'-sheaf.

Lemma 0.1. *With $\epsilon : (\mathcal{C}_\tau, \mathcal{O}_\tau) \to (\mathcal{C}_{\tau'}, \mathcal{O}_{\tau'})$ as above. Let A be a set and for $\alpha \in A$ let*

$$\begin{array}{ccc} E_\alpha & \longrightarrow & Y_\alpha \\ \downarrow & & \downarrow \\ Z_\alpha & \longrightarrow & X_\alpha \end{array}$$

be a commutative diagram in the category $\mathcal{C}$. Assume that

1. *a τ'-sheaf $\mathcal{F}'$ is a τ-sheaf if $\mathcal{F}'(X_\alpha) = \mathcal{F}'(Z_\alpha) \times_{\mathcal{F}'(E_\alpha)} \mathcal{F}'(Y_\alpha)$ for all α,*
2. *for K' in $D(\mathcal{O}_{\tau'})$ in the essential image of $R\epsilon_*$ the maps $c^{K'}_{X_\alpha, Z_\alpha, Y_\alpha, E_\alpha}$ of Lemma 0F16 are isomorphisms for all α.*

Then $K' \in D^+(\mathcal{O}_{\tau'})$ is in the essential image of $R\epsilon_*$ if and only if the maps $c^{K'}_{X_\alpha, Z_\alpha, Y_\alpha, E_\alpha}$ are isomorphisms for all α.

Proof. The "only if" direction is implied by assumption (2). On the other hand, if K' has a unique nonzero cohomology sheaf, then the "if" direction follows from assumption (1). In general we will use an induction argument to prove the "if" direction. Let us say an object K' of $D^+(\mathcal{O}_{\tau'})$ satisfies (P) if the maps $c^{K'}_{X_\alpha, Z_\alpha, Y_\alpha, E_\alpha}$ are isomorphisms for all $\alpha \in A$.

Namely, let K' be an object of $D^+(\mathcal{O}_{\tau'})$ satisfying (P). Choose a distinguished triangle

$$K' \to R\epsilon_*\epsilon^{-1}K' \to M' \to K'[1]$$

in $D^+(\mathcal{O}_{\tau'})$ where the first arrow is the adjuntion map. By (2) and Lemma 0EWP we see that M' has (P). On the other hand, applying ϵ^{-1} and using that $\epsilon^{-1}R\epsilon_* = \mathrm{id}$ by Section 0EWK we find that $\epsilon^{-1}M' = 0$. In the next paragraph we will show $M' = 0$ which finishes the proof.

Let K' be an object of $D^+(\mathcal{O}_{\tau'})$ satisfying (P) with $\epsilon^{-1}K' = 0$. We will show $K' = 0$. Namely, given $n \in \mathbf{Z}$ such that $H^i(K') = 0$ for $i < n$ we will show that $H^n(K') = 0$. For $\alpha \in A$ we have a distinguished triangle

$$R\Gamma_{\tau'}(X_\alpha, K') \to R\Gamma_{\tau'}(Z_\alpha, K') \oplus R\Gamma_{\tau'}(Y_\alpha, K') \to R\Gamma_{\tau'}(E_\alpha, K') \to R\Gamma_{\tau'}(X_\alpha, K')[1]$$

by Lemma 0F16. Taking cohomology in degree n and using the assumed vanishing of cohomology sheaves of K' we obtain an exact sequence

$$0 \to H^n_{\tau'}(X_\alpha, K') \to H^n_{\tau'}(Z_\alpha, K') \oplus H^n_{\tau'}(Y_\alpha, K') \to H^n_{\tau'}(E_\alpha, K')$$

which is the same as the exact sequence

$$0 \to \Gamma(X_\alpha, H^n(K')) \to \Gamma(Z_\alpha, H^n(K')) \oplus \Gamma(Y_\alpha, H^n(K')) \to \Gamma(E_\alpha, H^n(K'))$$

We conclude that $H^n(K')$ is a a τ-sheaf by assumption (1). However, since the τ-sheafification $\epsilon^{-1}H^n(K') = H^n(\epsilon^{-1}K')$ is 0 as $\epsilon^{-1}K' = 0$ we conclude that $H^n(K') = 0$ as desired. □